Asadullah Bin Rahman

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Research Interests: Image Processing, Machine Learning, Quantum Computing

Employment

07/25 – Present ◇ **Lecturer, Computer Science and Engineering**
World University of Bangladesh (WUB), Dhaka, Bangladesh
Roles: Undergraduate teaching, thesis & project supervision, competitive programming club mentoring, student counseling

07/23 – 07/25 ◇ **Research Assistant**
IoThink Lab, Department of Computer Science and Engineering, HSTU
Focus: Wavelet-based denoising and deep learning for medical image analysis; interdisciplinary dataset collection
Supervisor: Dr. Md. Abdulla Al Mamun

Education

07/23 – 07/25 ◇ **MSc in Computer Science and Engineering**, Hajee Mohammad Danesh Science and Technology University (HSTU), Bangladesh
Thesis: MRI Image Denoising using Wavelet Transform and Deep Learning Architectures
CGPA: 3.547/4.00

01/17 – 12/22 ◇ **BSc (Engineering) in Computer Science and Engineering**, Hajee Mohammad Danesh Science and Technology University (HSTU), Bangladesh
Thesis: MRI Brain Tumor Classification from Contrast-Enhanced MRI using Deep Learning
CGPA: 3.31/4.00

Skills

Quantum Computing	◇ Proficient in Qiskit, Cirq, PennyLane, Classiq; applied experience in Quantum Algorithms, Quantum Machine Learning, Quantum Key Distribution
Machine Learning	◇ Proficient in NumPy, Pandas, SciPy, Matplotlib, OpenCV, PyTorch, TensorFlow; applied experience in Image Processing, Computer Vision
Programming	◇ C++, Java, Python; Data Structures, Algorithms, Databases; Solved 300+ problems (CodeForces, HackerRank, UVa)
Languages	◇ Bangla (Native), English (IELTS Academic: **6.5**, CEFR Level: B2)
Others	◇ Linux, Git, LaTeX

Research Publications

Conference Proceedings

1. **A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, "Mitigating Noise from Biomedical Images Using Wavelet Transform Techniques," in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2025, pp. 1–6. DOI: 10.1109/ECCE64574.2025.11013062.

2. **A. B. Rahman**, M. Touhid Islam, M. R. Islam, M. Sohrawordi, and M. N. Sultan, "Enhanced Brain Tumor Classification from MRI Images Using Deep Learning Model," in *2023 26th International Conference on Computer and Information Technology (ICCIT)*, 2023, pp. 1–6. DOI: 10.1109/ICCIT60459.2023.10441064.

Hyperlinks are embedded in this CV. Please click anywhere with for details!

Preprints

1. **A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, *Deep Learning Architectures for Medical Image Denoising: A Comparative Study of CNN-DAE, CADTra, and DCMIEDNet*, 2025. arXiv: 2508.17223 [eess.IV]. URL: https://arxiv.org/abs/2508.17223.
2. **A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, *Systematic Evaluation of Wavelet-Based Denoising for MRI Brain Images: Optimal Configurations and Performance Benchmarks*, 2025. arXiv: 2508.15011 [eess.IV]. URL: https://arxiv.org/abs/2508.15011.

Teaching

Fall 2026
- ◇ Object Oriented Programming (Batch 71C, 72D, 71E, Semester 4). CSE 06132120.
- ◇ Introduction to Programming (Batch 74D, Semester 2). CSE 06131209.

Summer 2026
- ◇ Compiler Design (Batch 63A, Semester 11). CSE 06134156.
- ◇ Object Oriented Programming (Batch 70E, Semester 4). CSE 06132120.
- ◇ Introduction to Programming (Batch 73B, Semester 1). CSE 06131209.

Spring 2026
- ◇ Object Oriented Programming (Batch 69A, Semester 4). CSE 06132120.
- ◇ Introduction to Programming (Batch 70B, 71C, Semester 2). CSE 06131209.

Fall 2025
- ◇ Computer Fundamentals (Batch 71G, Semester 1). CSE 06111101.
- ◇ Object Oriented Programming (Batch 68A, Semester 4). CSE 06132120.
- ◇ Computer and Cyber Security (Batch 63B, Semester 9). CSE 06124158.

Thesis Supervision

Summer 2026
- ◇ **Explainable Deep Learning for Plant Disease Classification Using Convolutional Neural Networks @ WUB:** Comparison of CNN architectures for leaf disease classification on PlantVillage crops, with Grad-CAM saliency exposing the leaf regions behind each prediction.
- ◇ **ClassSync Pro: IoT-Based Smart Classroom Automation System @ WUB:** LMS-scheduled control of classroom air-conditioning, fans, lighting and projectors with sensor-based occupancy validation; projected 30–40% cut in campus energy use.

Coaching and Mentoring

07/25 – Present
- ◇ **Competitive Programming Coach @ WUB:** Training undergraduate teams for **ICPC** (Asia West Continent / Dhaka Regional), **NCPC** and inter-university programming contests (IUPC). Weekly problem-solving sessions and mock contests covering data structures, graph algorithms, dynamic programming, number theory and computational geometry; team formation, contest strategy and post-contest editorial review.

Research and Projects

02/26
- ◇ **Multilayered Security** : Implemented and simulated a research project on Multi-Layered Security System: Integrating Quantum Key Distribution with Classical Cryptography to Enhance Steganographic Security.

11/24
- ◇ **Guava Fruit Disease Dataset @ IoThink Lab** : Collaborated on dataset collection for interdisciplinary research.

Research and Projects (continued)

08/24 ◇ **Quantum Machine Learning Projects @ Womanium Program**: Four QML projects completed for *QML for Conspicuity Detection in Production.*
○ **QML for Anomaly Detection** 🔗 : hybrid quantum-classical model for anomaly detection.
○ **Quanvolutional Neural Networks** 🔗 : hybrid quantum convolutional model for MNIST digit classification.
○ **Quantum Variational Classifier** 🔗 : quantum classifier for penguin species classification.
○ **Quantum Regression Model** 🔗 : learned and predicted the sine function on the interval $[0, 2\pi]$.

Professional Development

09/24 – 12/24 ◇ **QClass24/25 Fall Semester** 🔗 **:** Achieved 96% in 3 ECTS graduate-level program on Quantum Algorithms and QKD.

06/24 – 08/24 ◇ **Womanium Quantum + AI 2024** 🔗 **:** Coursework on Quantum Computing and AI.

08/24 ◇ **QGSS 2024 – Quantum Excellence** 🔗 🔗 **:** Intensive quantum computing boot camp by IBM Qiskit.

◇ **Classiq Diploma** 🔗 **:** Advanced Quantum Algorithm design.

◇ **PennyLane Diploma** 🔗 **:** Quantum Machine Learning Challenge completion.

07/24 ◇ **QNickel Diploma** 🔗 **:** Quantum Algorithms workshop.

◇ **QBronze Diploma** 🔗 **:** Introductory Quantum Computing and Programming.

09/20 ◇ **Python for Everybody Specialization** 🔗 **:** Coursera specialization on Python fundamentals and data structures.

Awards and Achievements

06/25 ◇ **Unitary Hack 2025** 🔗 🔗 **:** For open source contributions to the quantum software ecosystem.

04/25 ◇ **YQuantum 2025** 🔗 🔗 **:** Solved BlueQubit's Peaked Circuits Challenge.

02/25 ◇ **MIT iQuHACK 2025** 🔗 **:** Solved IonQ's Max Cut problem.

08/24 ◇ **Womanium Quantum + AI 2024** 🔗 **:** Program finalist and QSL fellowship nominee.

06/24 ◇ **IBM Quantum Challenge 2024** 🔗 🔗 **:** Ranked 11th globally.

Volunteering Work

04/25 – Present ◇ **Coordinator & Mentor, QBangladesh (QWorld)** 🔗 **:** Mentoring students and promoting quantum computing literacy in Bangladesh.
Hosted Events: QBronze 🔗 , QSilver 🔗 .

Hobbies

◇ Chess 🔗

◇ Gardening